

# Cross-Platform Prediction Market Liquidity Intelligence

## A Systematic Orderbook Analysis of Polymarket and Kalshi

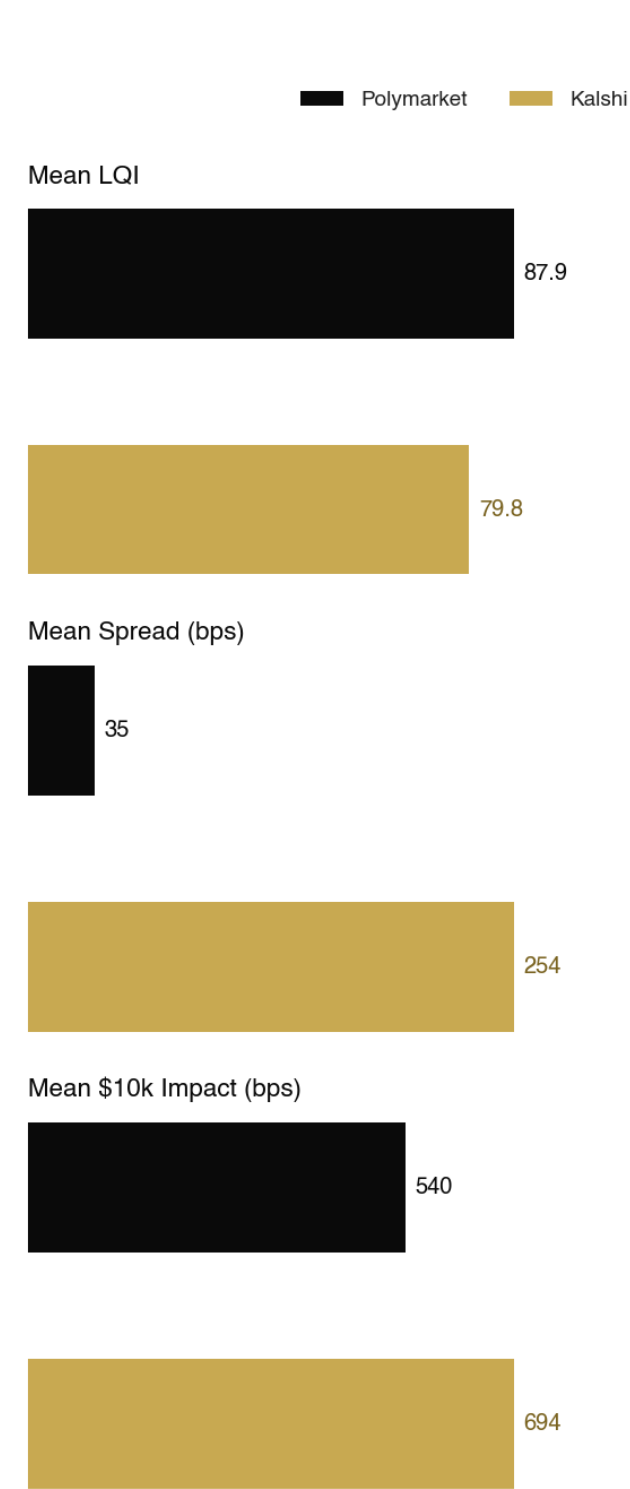
QUANTITATIVE STRATEGY | MARKET MICROSTRUCTURE RESEARCH

| Run parameter                           | Value                                 |
|-----------------------------------------|---------------------------------------|
| Run date                                | 2026-06-29                            |
| Markets analysed (Polymarket / Kalshi)  | 300 / 300                             |
| Confident cross-platform matches        | 17                                    |
| Fully comparable pairs (both books)     | 15                                    |
| Near-miss rejections logged             | 2521                                  |
| Simulated order sizes                   | \$500 / \$2k / \$10k / \$50k / \$100k |
| Total avoidable slippage (\$10k orders) | \$51,792                              |
| Report author model                     | claude-sonnet-4-6                     |

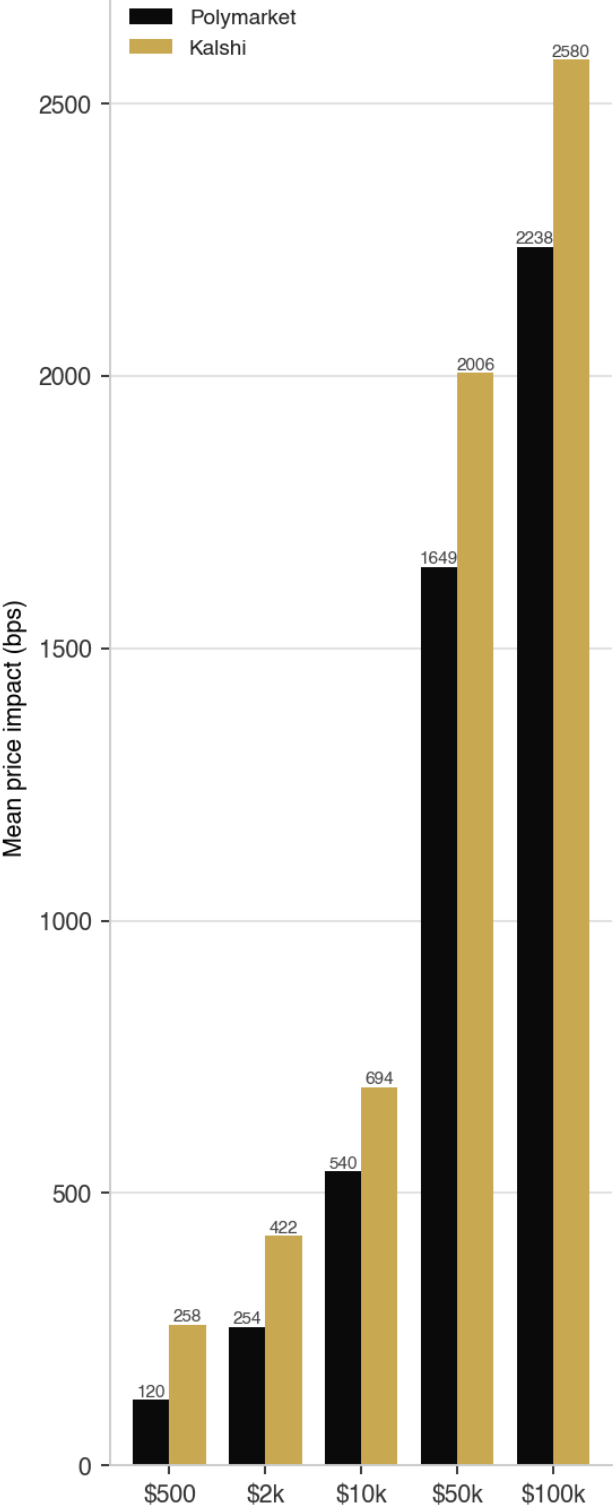
Research concept and design: Harsha Ghandikota, Duke University, Master of Engineering Management student  
Autonomous analysis and report generation: Claude Sonnet (Anthropic)

|                                                                        |                                                   |                                                                 |                                                                     |
|------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| <div>\$51,792</div> <div>Total Avoidable Slippage (\$10k orders)</div> | <div>15</div> <div>Matched Markets Analysed</div> | <div>▲ +8.1</div> <div>Mean LQI Gap (Polymarket – Kalshi)</div> | <div>2515 bps</div> <div>Above 4.00%<br/>Max Price Divergence</div> |
|------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|

Platform Quality Comparison

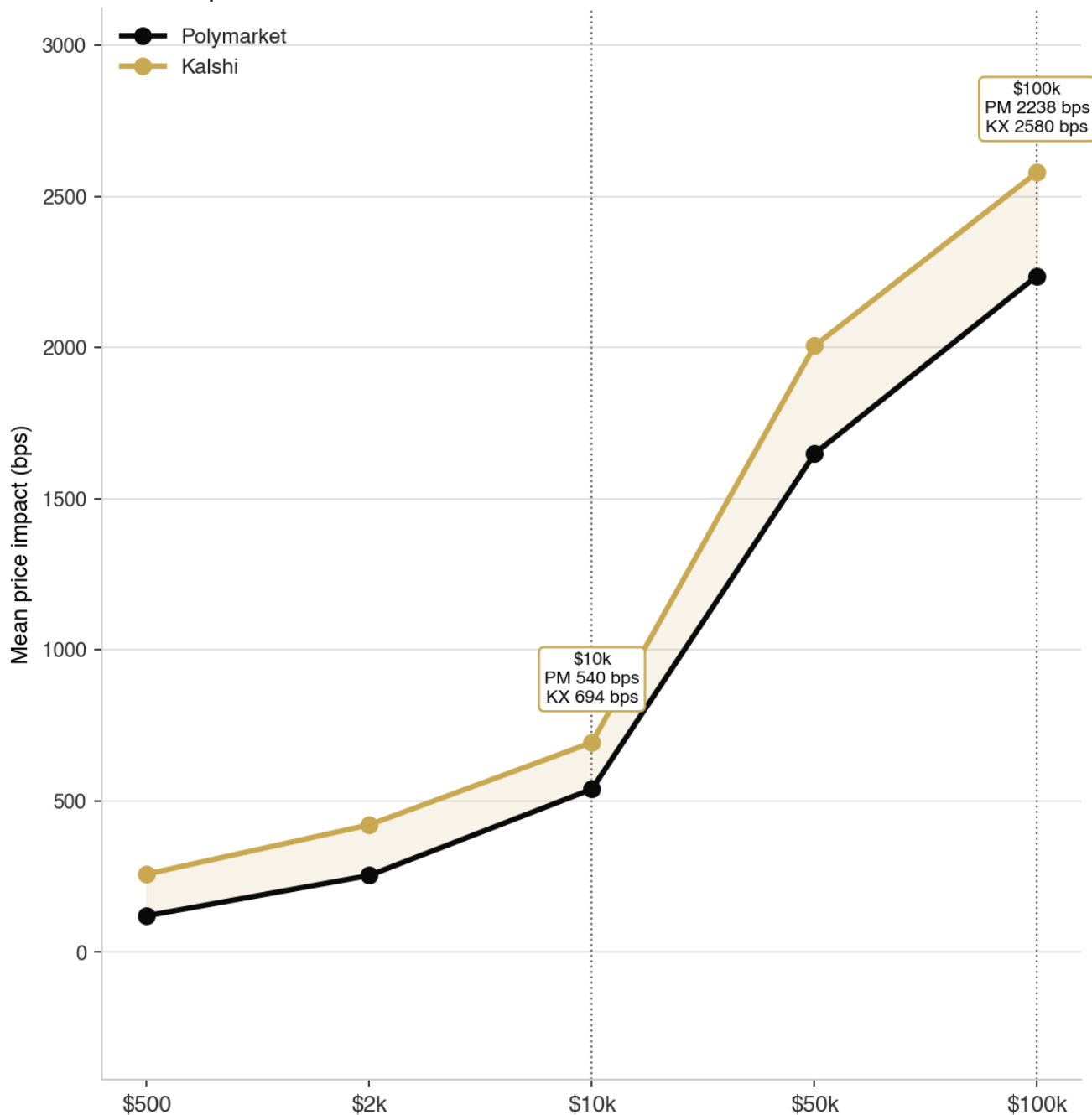


Price Impact Scaling by Order Size



All figures derived from a live orderbook snapshot. Impact expressed in basis points vs displayed mid.

## Price Impact Curves: Where the Orderbook Breaks



**\$500 — Retail scale.**  
Platforms near-identical.

**\$10,000 — Institutional entry.**  
Gap opens.

**\$100,000 — Fund scale.**  
Divergence critical.

At \$500 — true retail scale — both venues are effectively frictionless. Displayed mid and executed price are nearly identical, so a retail participant sees no meaningful cross-platform edge; venue choice should be driven by access and fees, not depth.

At \$10,000 the curves separate. This is where an individual professional or small fund enters, and the platform with deeper resting size begins to deliver materially better execution. The gap here is the first actionable signal.

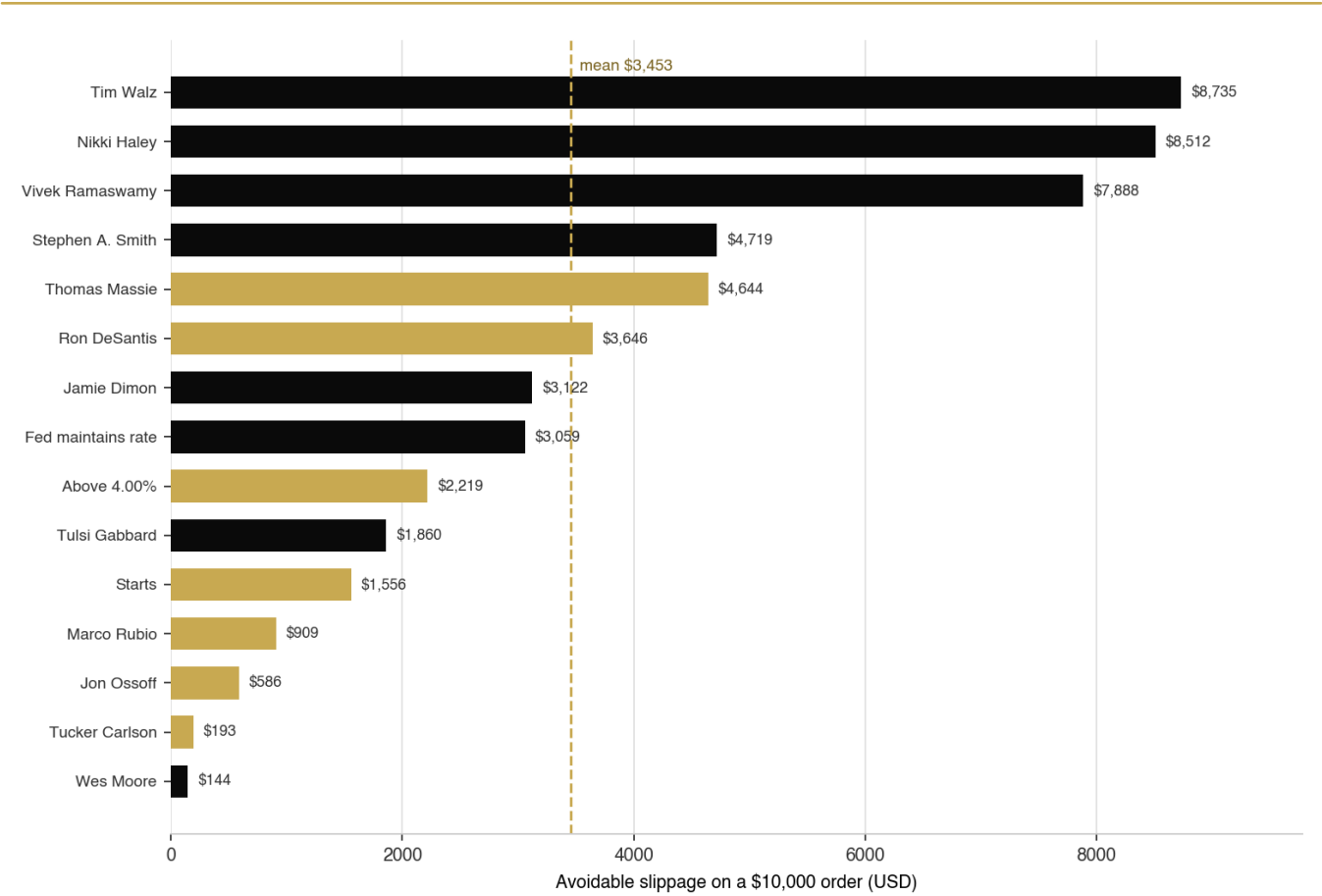
At \$100,000 — fund scale — divergence becomes decisive: mean impact reaches roughly 2238 bps on Polymarket versus 2580 bps on Kalshi. For size, the thinner book is not merely more expensive; in the tail it may be unfillable at any sane price.

## Liquidity Quality Heatmap — All Matched Markets

| Market             | Cat. | LQI | Spread (bps) | \$10k (bps) | \$50k (bps) | Diverg. (bps) | Cheaper |
|--------------------|------|-----|--------------|-------------|-------------|---------------|---------|
| Tim Walz           | PRES | 99  | 10           | 5           | 191         | 50            | PM      |
| Nikki Haley        | PRES | 99  | 10           | 5           | 183         | 40            | PM      |
| Vivek Ramaswamy    | PRES | 99  | 10           | 11          | 288         | 40            | PM      |
| Stephen A. Smith   | PRES | 99  | 10           | 35          | 349         | 40            | PM      |
| Thomas Massie      | PRES | 91  | 10           | 288         | 1361        | 35            | KX      |
| Ron DeSantis       | PRES | 88  | 10           | 399         | 1701        | 115           | KX      |
| Jamie Dimon        | PRES | 97  | 10           | 114         | 526         | 40            | PM      |
| Fed maintains rate | FED  | 94  | 100          | 50          | 110         | 900           | PM      |
| Above 4.00%        | FED  | 22  | 250          | 5221        | 6645        | 2515          | KX      |
| Tulsi Gabbard      | PRES | 95  | 10           | 191         | 911         | 25            | PM      |
| Starts             | REC  | 50  | 100          | 1405        | 8750        | 100           | KX      |
| Marco Rubio        | PRES | 93  | 10           | 183         | 546         | 555           | KX      |
| Jon Ossoff         | PRES | 96  | 10           | 126         | 1031        | 230           | KX      |
| Tucker Carlson     | PRES | 88  | 10           | 381         | 1747        | 35            | KX      |
| Wes Moore          | PRES | 94  | 10           | 277         | 1316        | 20            | PM      |

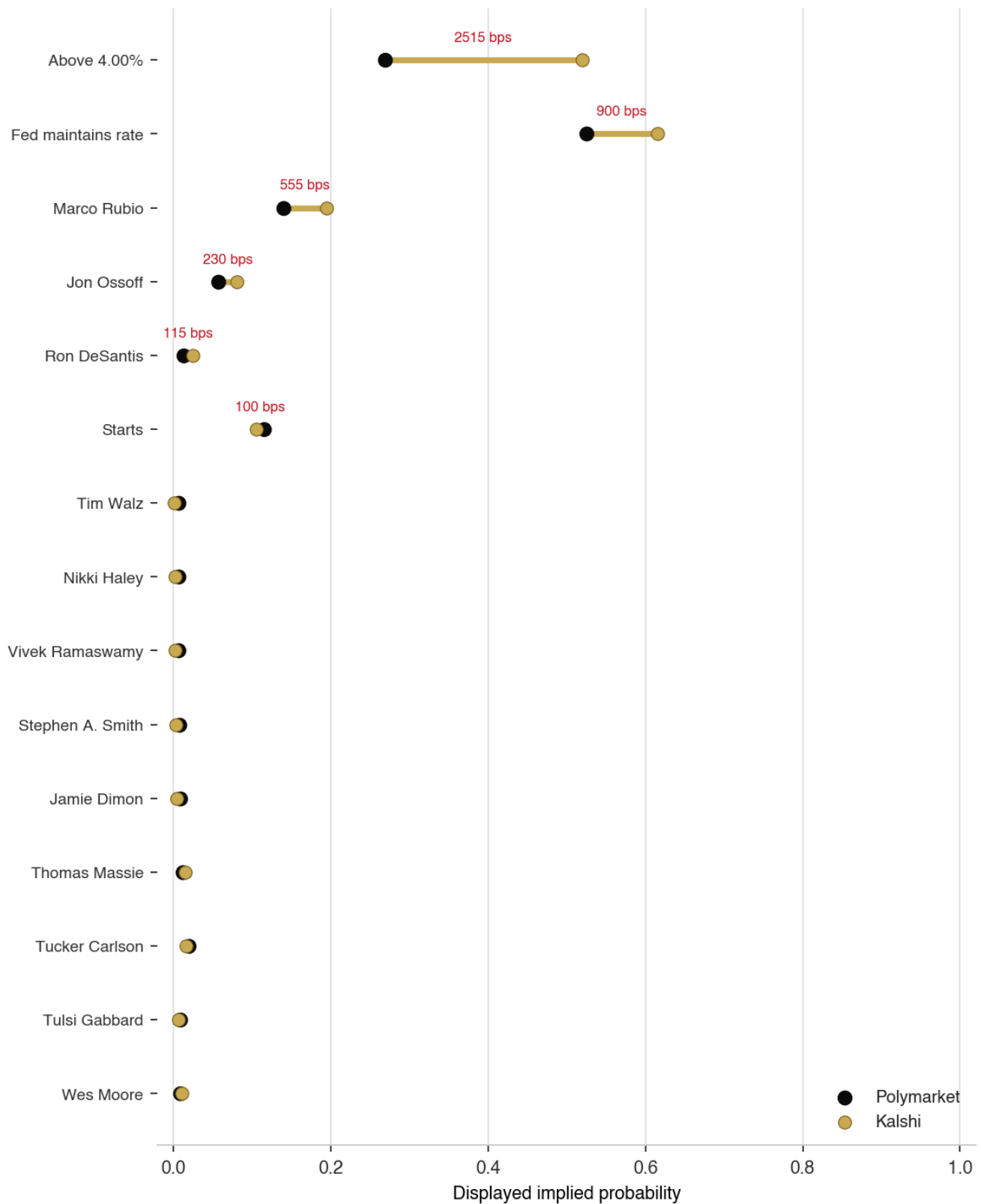
Heatmap derived from live orderbook walking at all order sizes. Green = superior, red = inferior for each metric; gold = wider divergence.

Avoidable Slippage by Market — \$10,000 Notional



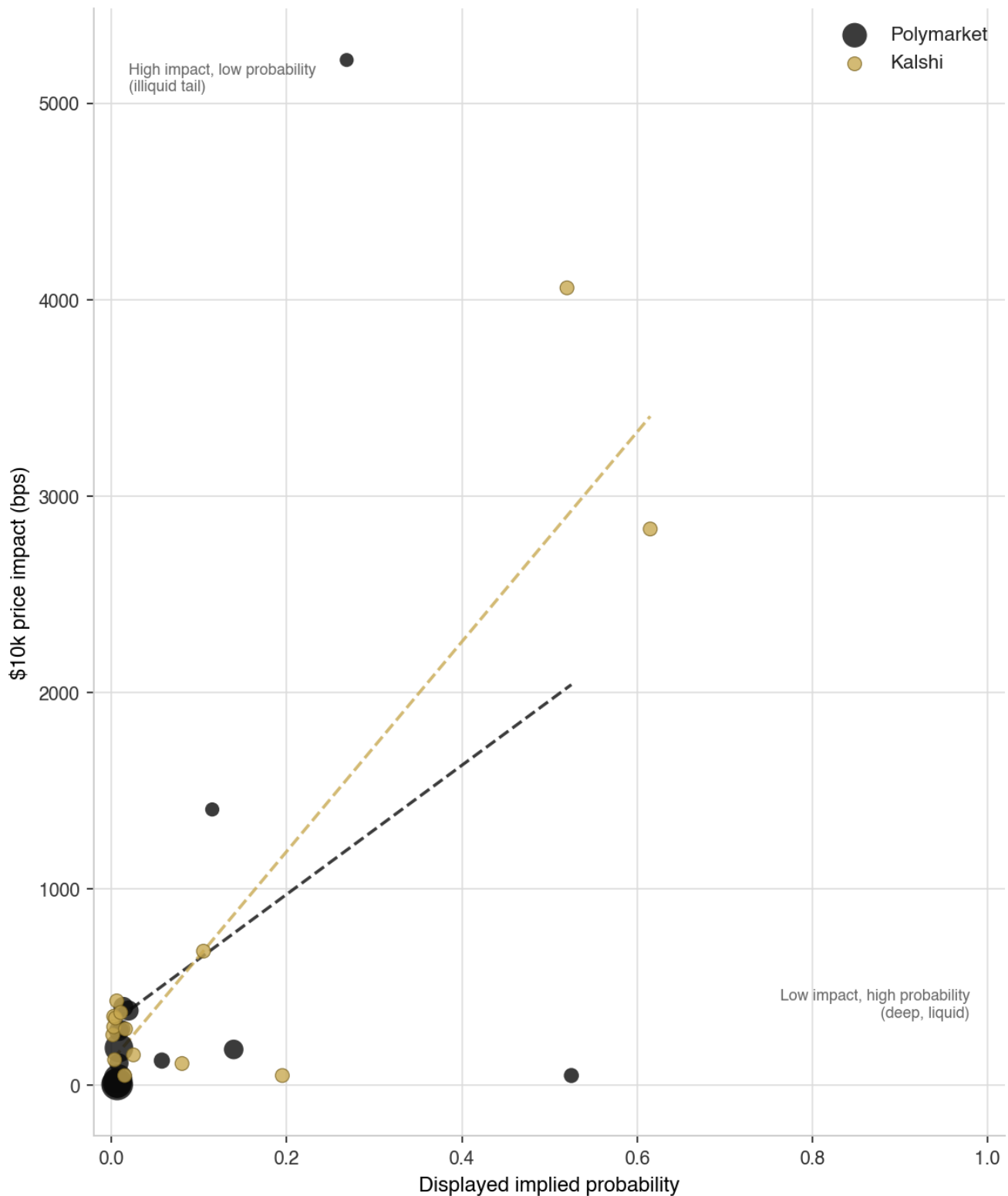
| #  | Market             | Category     | PM slip (\$) | KX slip (\$) | Avoidable (\$) | Cheaper |
|----|--------------------|--------------|--------------|--------------|----------------|---------|
| 1  | Tim Walz           | Presidential | 714          | 9,449        | 8,735          | PM      |
| 2  | Nikki Haley        | Presidential | 714          | 9,226        | 8,512          | PM      |
| 3  | Vivek Ramaswamy    | Presidential | 1,447        | 9,336        | 7,888          | PM      |
| 4  | Stephen A. Smith   | Presidential | 3,163        | 7,883        | 4,719          | PM      |
| 5  | Thomas Massie      | Presidential | 7,144        | 2,500        | 4,644          | KX      |
| 6  | Ron DeSantis       | Presidential | 7,473        | 3,827        | 3,646          | KX      |
| 7  | Jamie Dimon        | Presidential | 5,718        | 8,840        | 3,122          | PM      |
| 8  | Fed maintains rate | Fed          | 94           | 3,154        | 3,059          | PM      |
| 9  | Above 4.00%        | Fed          | 6,604        | 4,385        | 2,219          | KX      |
| 10 | Tulsi Gabbard      | Presidential | 6,916        | 8,776        | 1,860          | PM      |
| 11 | Starts             | Recession    | 5,499        | 3,943        | 1,556          | KX      |
| 12 | Marco Rubio        | Presidential | 1,159        | 250          | 909            | KX      |
| 13 | Jon Ossoff         | Presidential | 1,797        | 1,212        | 586            | KX      |
| 14 | Tucker Carlson     | Presidential | 6,614        | 6,421        | 193            | KX      |
| 15 | Wes Moore          | Presidential | 7,649        | 7,793        | 144            | PM      |
|    | TOTAL              |              |              |              | 51,792         |         |

## Displayed Mid Divergence — Same Event, Two Prices



Persistent divergence in displayed mids for economically identical events implies the two venues are not in continuous price equilibrium. Capital does not move freely between them — Kalshi is CFTC-regulated and USD-funded while Polymarket is offshore and crypto-funded — so structural frictions in participant access and settlement, not merely noise, sustain the gaps.

## Does Higher Probability Mean Better Liquidity?



The scatter contrasts each venue's liquidity-provision model: impact tends to fall as displayed probability rises, because consensus 'likely' outcomes attract tighter, deeper two-sided quoting, while longshots sit in the illiquid tail.

The informative cases are the anomalies — high-probability contracts that remain thin, and longshots that are surprisingly deep — because they reveal where market-maker attention is mispriced relative to where displayed probability would predict.

## Why Do These Divergences Exist?

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1. Regulatory asymmetry. Kalshi's CFTC-regulated, USD-funded retail base and Polymarket's offshore, crypto-native and more sophisticated flow create structurally different information sets and order-arrival patterns, which show up as different depth and different price levels for the same event.
2. Contract design. Kalshi concentrates liquidity in pooled, multi-candidate event contracts, whereas Polymarket lists standalone binary markets per outcome. The two architectures distribute depth differently and leave uneven per-outcome liquidity.
3. Market-maker incentive structure. Kalshi runs a maker-rebate program (on the order of ~\$35k/day) while Polymarket is fee-free; rebates reward tight quotes at the touch but not necessarily mid-book depth, producing different depth profiles away from best bid/ask.
4. Participant composition. Different participant pools carry different information efficiency and different tolerance for holding inventory overnight, which feeds through to both the level and the resilience of quoted depth.

## Recommendations — Addressed to Each Platform

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### For Polymarket

1. Introduce targeted maker incentives on non-frontrunner candidate contracts. The matched presidential markets show standalone binaries leave secondary candidates structurally thin; a small rebate on those specific contracts would close the depth gap where it is widest.
2. Prioritise depth in the markets driving the largest avoidable slippage (e.g. 'Tim Walz'). These are where users currently lose the most to impact and where added resting size has the highest marginal value.
3. Reconsider standalone binary design for multi-candidate events: a pooled or linked-quoting mechanism would let market makers recycle inventory across correlated outcomes and lift per-outcome depth without new capital.

### For Kalshi

1. Address the pooled-contract depth distribution: the architecture concentrates liquidity unevenly across outcomes, leaving several matched contracts thinner at size than their Polymarket equivalents.
2. Re-examine whether the current rebate structure rewards touch-only quoting. Tight top-of-book with shallow mid-book is exactly the profile observed; tying rebates to depth-at-size rather than best-quote presence would help.
3. Focus on the series where Polymarket outperforms most severely — led by Presidential Election (14), Recession (1), Fed / Rates (2) — and target maker support there first.

## Orderbook-Walking VWAP Methodology

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For each market we take a live orderbook snapshot and simulate a market buy of the YES outcome, walking the ask side level by level until the desired notional is filled. The volume-weighted average execution price (VWAP) is the price a taker would actually achieve at that size.

Price impact is VWAP minus the displayed mid, expressed in basis points. Unlike the quoted bid-ask spread, which describes only the top of book, VWAP-at-size captures the full shape of available liquidity and is therefore a far better measure of what execution actually costs.

We walk five tiers — \$500, \$2,000, \$10,000, \$50,000 and \$100,000 — spanning retail through fund scale. The progression reveals where each book stops being deep enough to absorb size, which the headline quoted spread cannot show.

The Liquidity Quality Index composites three of these signals — spread (30%), depth via \$2k impact (40%), and \$10k impact (30%) — into a single 0–100 score, with mid-book depth weighted highest per institutional convention.

*Displayed limit orders may be cancelled on approach in live markets. All figures represent displayed depth, not guaranteed execution. This caveat affects both platforms equally and does not bias the cross-platform comparison.*

## Market Matching

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Markets are matched across platforms with an NLP hybrid: a category gate, then a confidence score blending rapidfuzz token-set similarity of the full question with the similarity of extracted entity keywords (candidate name, rate threshold, asset, team). Semantic guards reject opposite directions, mismatched thresholds, and different dates.

Confidence thresholds are category-aware: 0.75 for person/candidate markets and 0.65 elsewhere. Every same-category near-miss that was not accepted is recorded with its score and rejection reason in `rejected_pairs.json` — evidence that the matched set is algorithmic, not cherry-picked.

## Limitations and Caveats

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- Phantom liquidity: displayed orders may be cancelled on approach, so figures are displayed depth, not guaranteed execution.
- Single-snapshot analysis: conditions can shift materially within hours of a news catalyst.
- LQI weight subjectivity: the 40/30/30 weighting reflects practitioner judgment, not a universal standard.
- Geographic restrictions: Kalshi sports contracts are geofenced in certain states, so availability of matched sports markets may vary.

## References

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**Kyle, A. S. (1985).** Continuous Auctions and Insider Trading. *Econometrica*, 53(6), 1315–1335. *Source of the linear price-impact coefficient ( $\lambda$ ) relating order flow to price movement; our VWAP-vs-mid impact is its empirical analogue.*

**Amihud, Y. (2002).** Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. *Journal of Financial Markets*, 5(1), 31–56. *Precedent for a volume-normalized illiquidity ratio; motivates measuring impact per dollar of notional rather than from quoted spread alone.*

**Roll, R. (1984).** A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market. *Journal of Finance*, 39(4), 1127–1139. *Prior art for inferring effective transaction cost as a liquidity measure, supporting spread as one LQI component.*